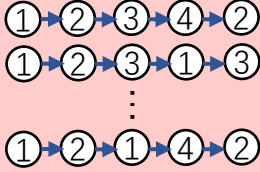


Random walk: $l = 5$



$$e_v = (f_v(\tilde{p}_1), f_v(\tilde{p}_2), \dots, f_v(\tilde{p}_M))$$



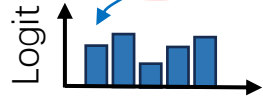
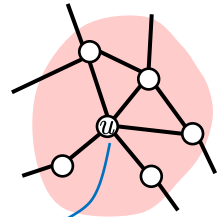
$$S^{(AP)}(u, v) = \frac{e_u \cdot e_v}{\|e_u\|_2 \|e_v\|_2}$$

s.t. dist(u, v) ≤ k

**Anonymous-Path-Driven
Graph Rewiring**



$$T = \text{softmax}(\text{SGC}(A, X; \Omega))$$



$$S^{(CA)}(u, v) = \frac{t_u \cdot t_v}{\|t_u\|_2 \|t_v\|_2}$$

s.t. dist(u, v) ≤ k

**Class-Affinity-Driven
Graph Rewiring**



$$\omega_i^{(AP)} = \mathbf{q}^\top \tanh(W_{AP} S_i^{(AP)})$$

$$\omega_i^{(CA)} = \mathbf{q}^\top \tanh(W_{CA} S_i^{(CA)})$$

$$\alpha_i^{(AP)}, \alpha_i^{(CA)} = \text{softmax}([\omega_i^{(AP)}, \omega_i^{(CA)}])$$

Self-Attention-Based Graph Fusion

$$S_i = \alpha_i^{(AP)} S^{(AP)} + \alpha_i^{(CA)} S^{(CA)}$$



Optimization Objective:

$$\mathcal{L}_{\text{seesaw}} = L_{\text{GNN}_S} + \lambda L_{\text{GNN}_A}$$

Incremental Trick: $S^* \leftarrow S_{\text{new}}$

Masking Trick: $S_{\text{masked}} = S \odot \mathbb{I}_{\geq \tau}$

Graph Learning on the Balanced Graph Topology